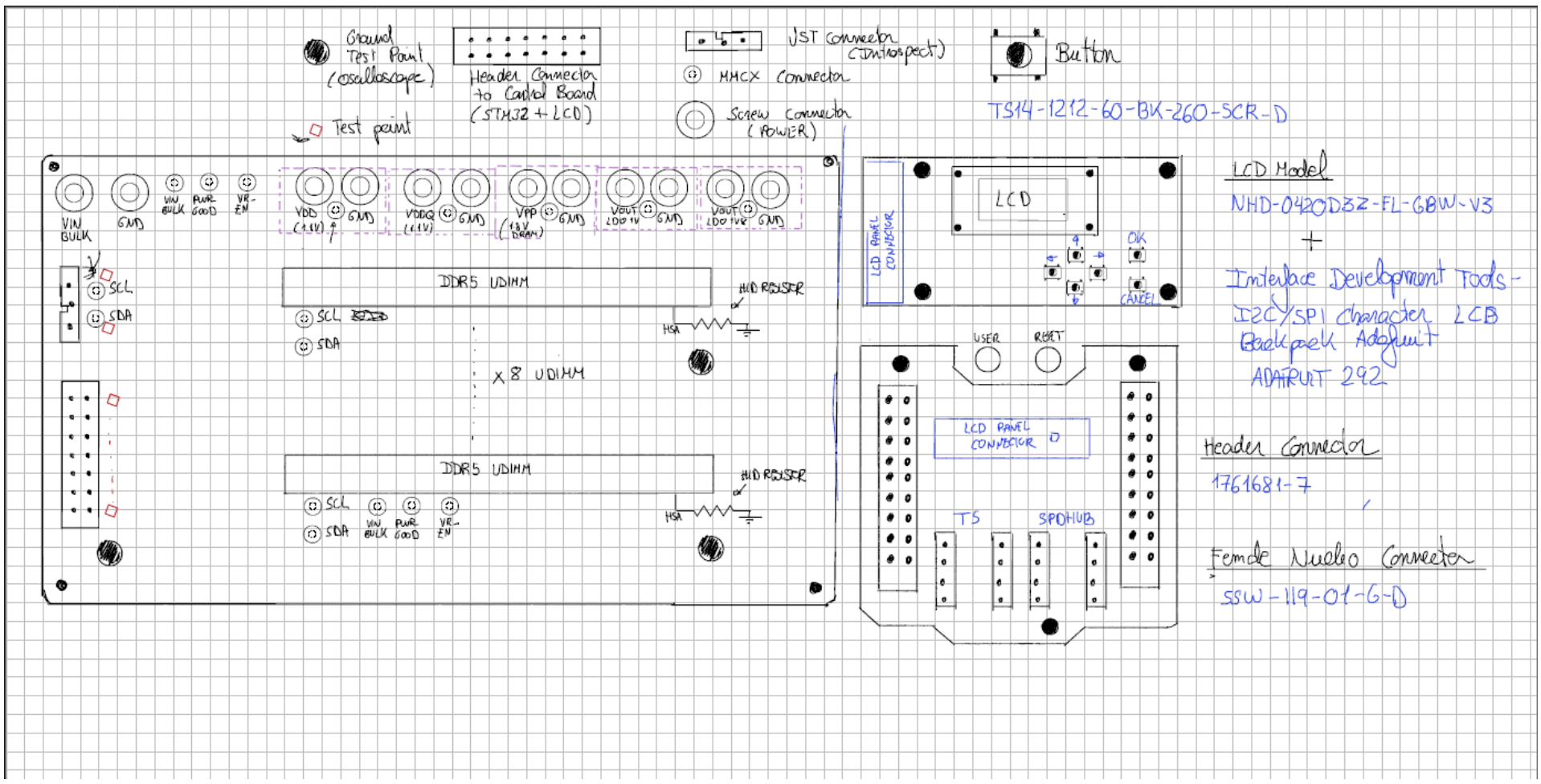
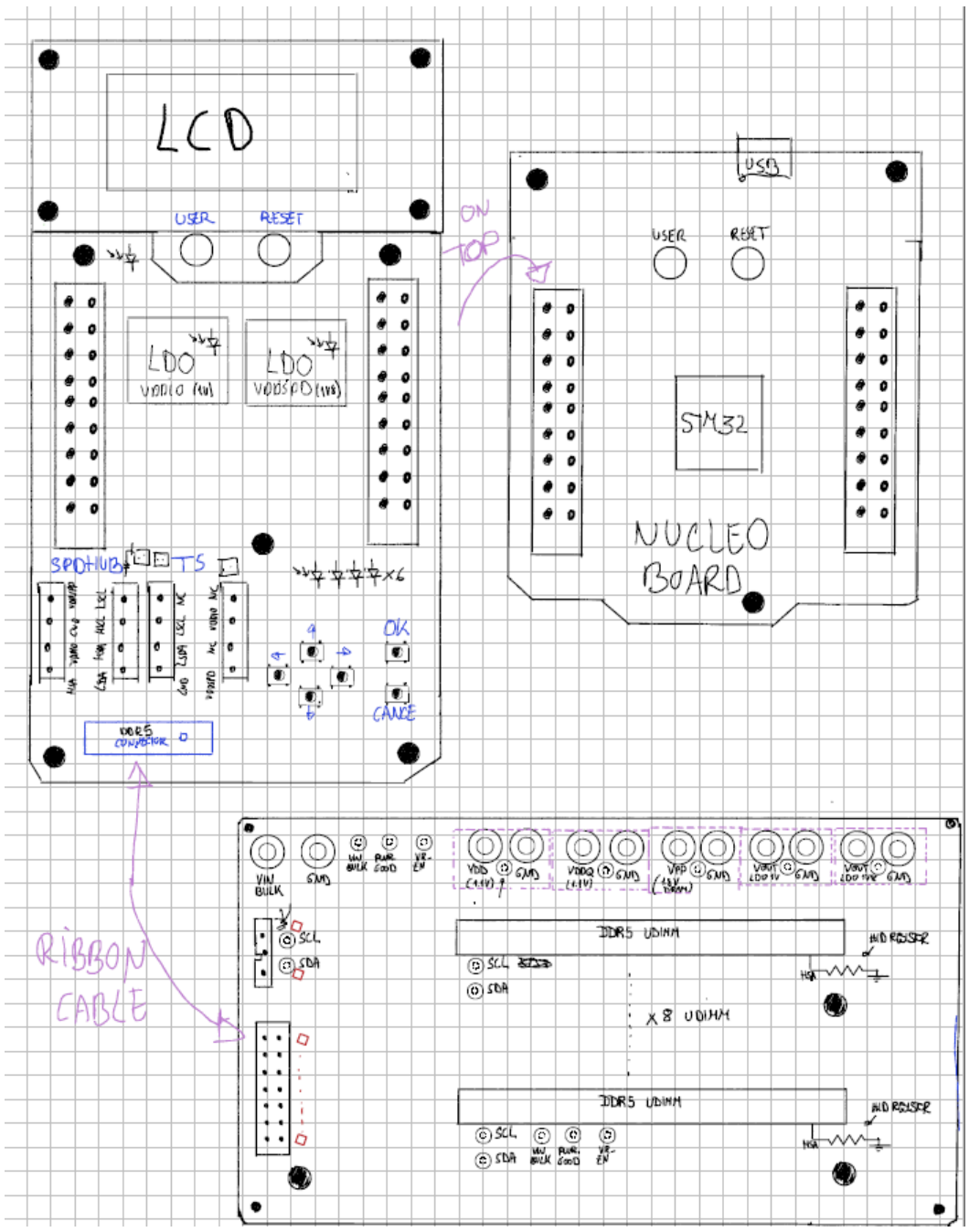
DDR5 test Platform

Open DDR5 test Platform

Develop a DDR5 test platform based on an STM32H563ZI





Capable of:

- Identifying and validating DDR5 modules
- Performing minimal but credible functional tests
- Displaying clear results (PASS / FAIL / WARN)
- Generating logs for analysis and traceability

Not included:

- Accessing the high-speed DDR bus

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- Emulating a DDR controller

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2. Microcontroller Peripherals and Their Contribution to Testing

I2C/I3C (DDR5 management channel)

Usage

- Communication with SPD Hub, TS, and PMIC

Contribution

- Module identification
- Reading status, telemetry, and configuration
- Foundation for all tests

GPIO + EXTI

Usage

- Module presence detection
- POWER_EN control
- Reading PGOOD / FAULT
- User buttons

Contribution

- Safe module control

ADC + DMA

Usage

- Measuring module supply lines (V/I)

Contribution

- Actual PMIC validation
- UV/OV detection
- Increases test credibility (“not only what the PMIC reports”)

Effort

- Medium (dividers, filtering, calibration)

SPI (LCD)

Usage

- Test selection menu
- Progress and results display

Contribution

- Clear, standalone interface
- Bench use without a PC

UART / USB CDC

Usage

- Logging to PC
- Debugging and exporting results

Contribution

- Traceability

- Offline analysis

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Usage

- Automatic recovery in case of lockup

Contribution

- Robustness for hot-plug and bus errors

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3. DDR5 Test Platform PCB

Included in the MVP

- High-quality DDR5 socket
- 1.8 V I2C channel with:
  - Proper pull-ups
  - ESD protection
  - Optional series resistor
- Power control:
  - Line enables
  - Test points
  - ADC measurement
- Auxiliary signals:
  - Presence
  - Fault / PGOOD
- Solid ground plane
- Power / analog / digital separation

Intentionally excluded from the MVP

- Routing DDR signals (DQ, DQS, CK)
- DDR controller
- SI simulations
- Length matching

The board is not a functional DDR5 board, but a test platform for management and power.

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4. Tests Included in the MVP

Platform Tests

- Power self-test
- I2C bus sanity check
- Module detection

SPD Hub

- Device scanning
- Basic SPD readout
- Full dump + CRC/check
- Consistency verification

Thermal Sensor (TS)

- Temperature reading
- Stability
- Reasonable range

PMIC

- Identification (if available)
- Error flag readout
- Basic telemetry
- Power ON/OFF sequencing

- Validation with external measurement

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- Communication bus stress
- Bus recovery
- Controlled hot-plug

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5. Critical Points to Meet the 4-Month Timeline

- Correct handling of hot-plug
- Recovery from I2C bus lockup
- Separating read-only tests from intrusive ones
- Restoring registers after write tests
- Logging from day one

---

6. Advanced Features (Outside the MVP)

For a second phase:

- Fault injection (UV/OV/OC)
- Controlled electronic loads
- Full I3C support
- Module database
- PC GUI
- Dynamic thermal tests

---

7. Conclusion

I await confirmation to:

1. Finalize the definitive MVP scope
2. Start schematic/PCB development
3. Define the firmware development roadmap

Memories Pinout (RDIMM / UDIMM / SODIMM)

Memories_Pinout.pdf

8. Pin Assignment

stm32h563zi Pinout	NUCLEO-H563ZI Pinout

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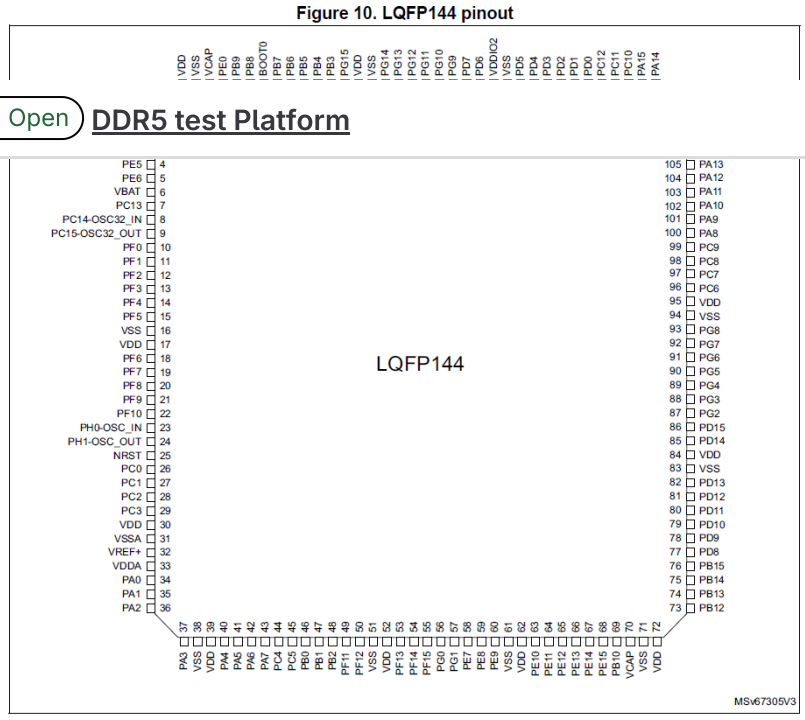
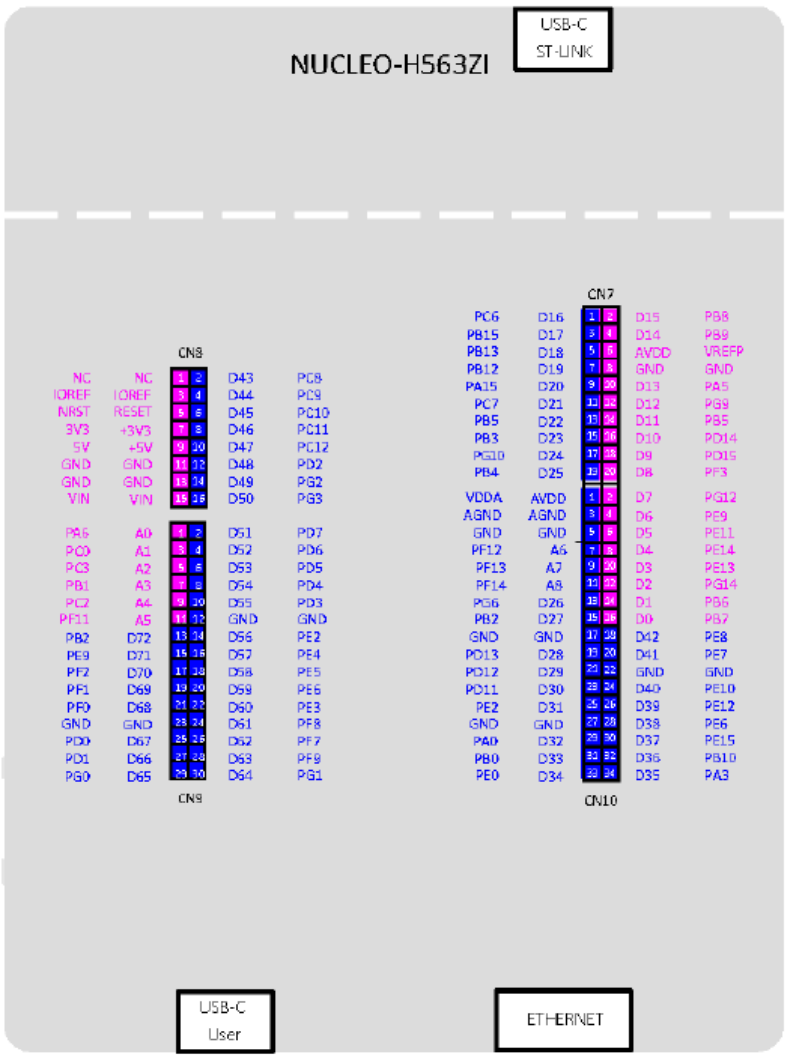
Figure 10. LQFP144 pinout  1. The above figure shows the package top view.	Figure 17. STM32H5 Nucleo-144 board 
STM32H563ZI | Product - STMicroelectronics	STM32H5 Nucleo-144 board (MB1404) - User manual
Schematic: MB1404	**I3C Manual:** https://www.st.com/resource/en/application_note/an5879-introduction-to-i3c-for-stm32-mcus-stmicroelectronics.pdf

GPIOs

ID	Pin_Name	Function	Microcontroller_Pin	Connected to DDR5 Test Board	
1	**PWR_GOOD_RDIMM_0**	Read Status PG RDIMM0	PC8	Yes	**GPIO_IN**
2	**PWR_GOOD_RDIMM_1**	Read Status PG RDIMM1	PC9	Yes	**GPIO_IN**

3	**PWR_GOOD_RDIMM_2**	Read Status PG RDIMM2	PC10	Yes	**GPIO_IN**

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		RDIMM3			
5	**PWR_GOOD_RDIMM_4**	Read Status PG RDIMM4	PC12	Yes	**GPIO_IN**
6	**PWR_GOOD_RDIMM_5**	Read Status PG RDIMM5	PD2	Yes	**GPIO_IN**
7	**PWR_GOOD_RDIMM_6**	Read Status PG RDIMM6	PG2	Yes	**GPIO_IN**
8	**PWR_GOOD_RDIMM_7**	Read Status PG RDIMM7	PG3	Yes	**GPIO_IN**
9	**PWR_ENABLE_RDIMM_0**	Enable/Disable PMIC RDIMM0	PD7	Yes	**GPIO_OUT**
10	**PWR_ENABLE_RDIMM_1**	Enable/Disable PMIC RDIMM1	PD6	Yes	**GPIO_OUT**
11	**PWR_ENABLE_RDIMM_2**	Enable/Disable PMIC RDIMM2	PA6	Yes	**GPIO_OUT**
12	**PWR_ENABLE_RDIMM_3**	Enable/Disable PMIC RDIMM3	PC0	Yes	**GPIO_OUT**
13	**PWR_ENABLE_RDIMM_4**	Enable/Disable PMIC RDIMM4	PC3	Yes	**GPIO_OUT**
14	**PWR_ENABLE_RDIMM_5**	Enable/Disable PMIC RDIMM5	PB1	Yes	**GPIO_OUT**
15	**PWR_ENABLE_RDIMM_6**	Enable/Disable PMIC RDIMM6	PD5	Yes	**GPIO_OUT**
16	**PWR_ENABLE_RDIMM_7**	Enable/Disable PMIC RDIMM7	PD4	Yes	**GPIO_OUT**
17	**LCD_UP_ARROW**	Button Up LCD Menu	PD3 (INTERRUPT)	No	**GPIO_IN**
18	**LCD_DOWN_ARROW**	Button Down LCD Menu	PF2 (INTERRUPT)	No	**GPIO_IN**
19	**LCD_LEFT_ARROW**	Button Left LCD Menu	PF8 (INTERRUPT)	No	**GPIO_IN**
20	**LCD_RIGHT_ARROW**	Button Right LCD Menu	PD0 (INTERRUPT)	No	**GPIO_IN**
21	**LCD_OK_BUTTON**	Button OK LCD Menu	PF7 (INTERRUPT)	No	**GPIO_IN**
22	**LCD_CANCEL_BUTTON**	Button CANCEL LCD Menu	PD1 (INTERRUPT)	No	**GPIO_IN**
23	**TRIGGER_IN_0**	Trigger Signal Input for Test	PF9 (INTERRUPT)	Yes	**GPIO_IN**

24	**TRIGGER_IN_1**	Trigger Signal Input for Test	PG0	Yes	**GPIO_IN**

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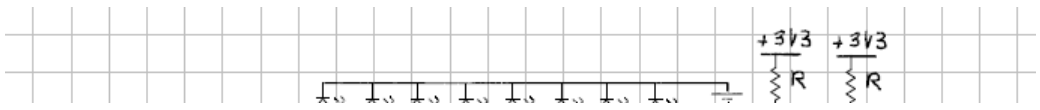
25	**TRIGGER_OUT_0**	Trigger Signal Output for Test	PG1 (TIM)	Yes	**GPIO_OUT**
26	**TRIGGER_OUT_1**	Trigger Signal Output for Test	PE0 (TIM)	Yes	**GPIO_OUT**
27	**GPIO_AUX_0**	GPIO Aux for Debug	PA3	Yes	**GPIO_IN / GPIO_OUT**
28	**GPIO_AUX_1**	GPIO Aux for Debug	PB0	Yes	**GPIO_IN / GPIO_OUT**
29	**CLK_LED_STATUS**	Clock Led Shift Register	PB10	Yes	**GPIO_OUT**
30	**DATA_LED_STATUS**	Data Led Shift Register	PE15	Yes	**GPIO_OUT**
31	**LATCH_LED_STATUS**	Latch Shift Register	PE12	Yes	**GPIO_OUT**
32	**RESET_LED_STATUS**	Reset Shift Register	PE10	Yes	**GPIO_OUT**

Communications

ID	Pin_Name	Function	NUCLEO_Pin	Connected to DDR5 Test Board
1	**Debug_UART_TX**	Data Logging and Debug Test Transmission Line	**USART3_TX** (PD8)	No
2	**Debug_UART_RX**	Data Logging and Debug Test Reception Line	**USART3_TX** (PD9)	No
3	**LCD_I2C_SCL**	Clock Signal I2C	**I2C2_SCL** (PF1)	No
4	**LCD_I2C_SDA**	Data Signal I2C	**I2C2_SDA** (PF0)	No
5	**I3C_SideBand_SCL**	Clock Signal I3C	**I3C1_SCL** (PB8)	Yes
6	**I3C_SideBand_SDA**	Data Signal I3C	**I3C1_SDA** (PB9)	Yes

LED Control

https://www.ti.com/lit/ds/symlink/sn54hc595.pdf?ts=1771422426933&ref_url=https%253A%252F%252Fwww.mouser.fr%252F



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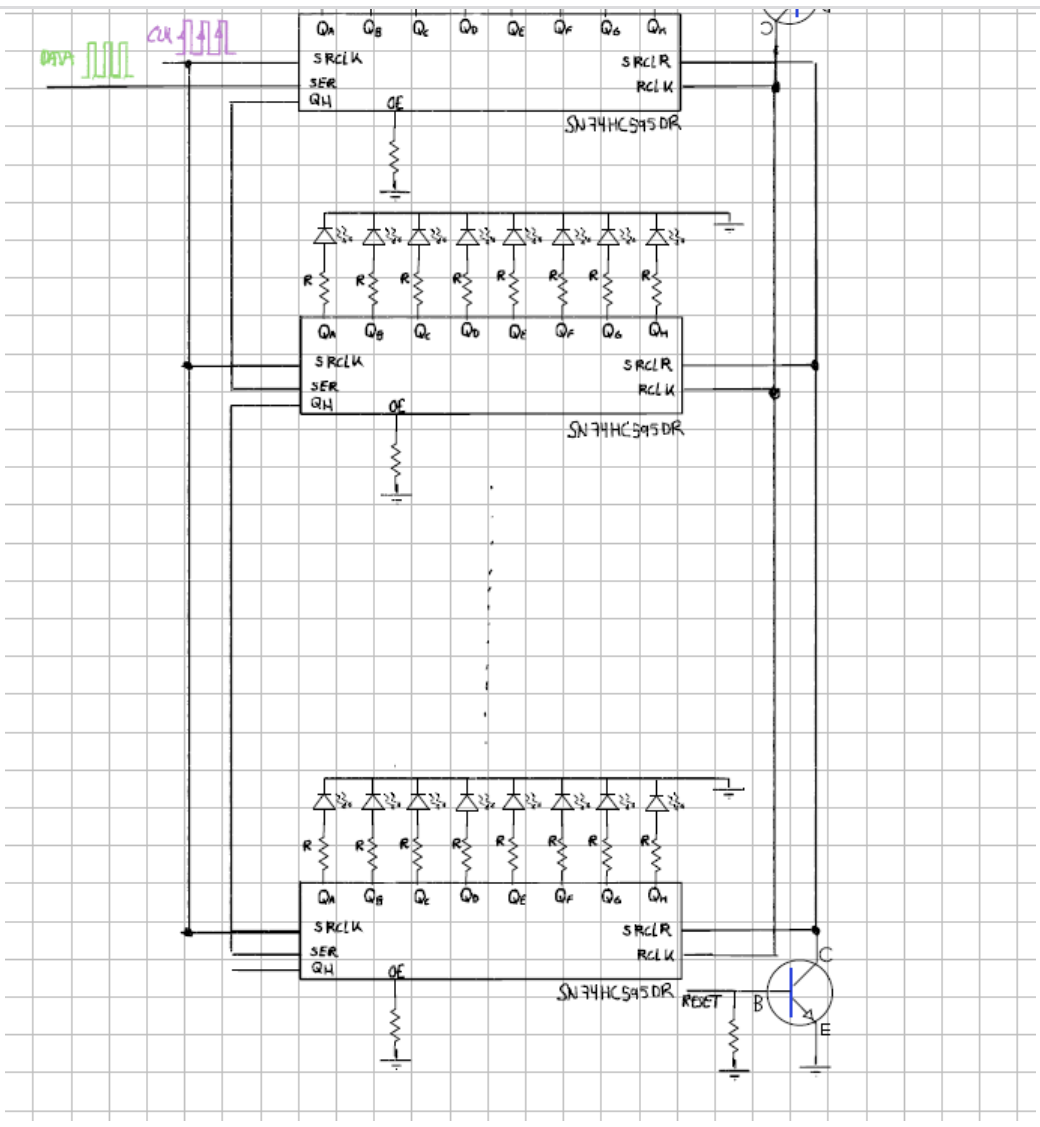

Peripheral	Pin Quantity
GPIO	32
I3C	2
I2C	2
Total	36

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Drag your designs here or click to upload.

No child items are currently assigned. Use child items to break down this issue into smaller parts.

Link issues together to show that they're related. Learn more.

Activity

- Xavier Rocher assigned to @Xavier.Rocher 6 days ago
- Xavier Rocher changed the description 6 days ago

X Xavier Rocher · 6 days ago Author Developer

@BILL.Snyder @Roberto.Jardim @Miguel.Inocencio @Leonardo.Sala @Pedro.Silva

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- Xavier Rocher changed the description 6 days ago

Leonardo Sala · 5 days ago Reporter

With the plans for this development, should we include basic checks of all of our Dimm components?

1. SPD

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3. PMIC
4. RCD
5. CKD

What about being able to read/write the device specific registers if debugging?

Bill

Leonardo Sala · 5 days ago Reporter

Hi Bill,

I would focus on DIMM-related potential issues, like:

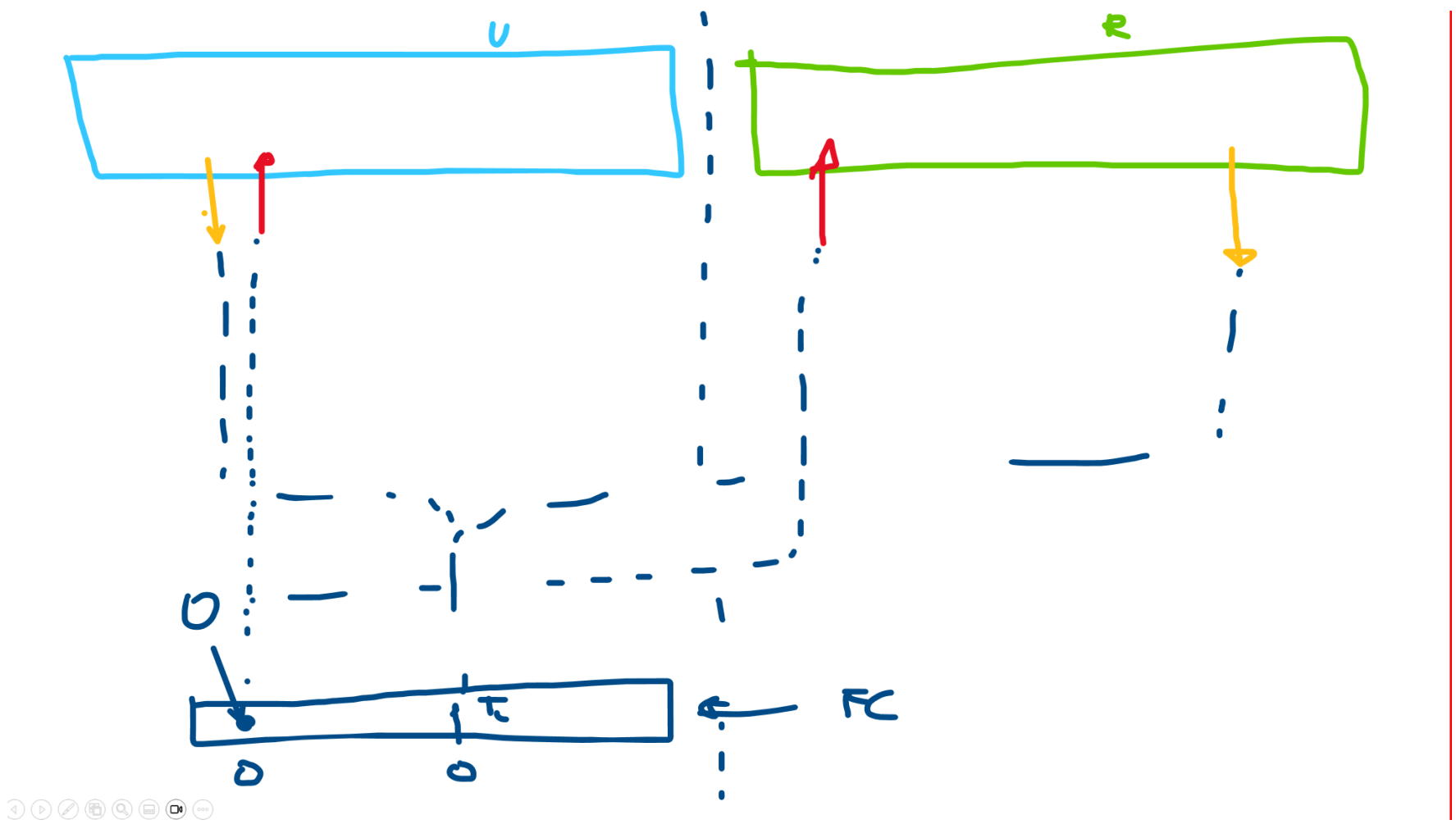
1. SPD memory parsing and cross-checking with Vendor ID/Version
2. SPD Memory CRC Checking
3. At speed testing of various/all protocols with every single component on the DIMM (e.g. R/W with IBI Header and PEC)
4. Pseudo Boot Sequence:
    1. SPD Parsing
    2. SETHID
    3. SETAASA
    4. TS reading
    5. ???

Reply...

Xavier Rocher changed the description 1 day ago

Xavier Rocher changed the description 1 day ago

Leonardo Sala · 1 minute ago Reporter



Leonardo Sala · just now Reporter

2 main checks:

1. Is 16 large enough number (8 inputs, 8 outputs) ?
2. Is the voltage range compatible for UDIMM/RDIMM/SODIMM ?

Xavier Rocher · just now Author Developer

Is the voltage range compatible for UDIMM/RDIMM/SODIMM ?

UDIMM and SODIMM are compatible (**VIN_Bulk:** 5V), RDIMM has two supplies (**VIN_Bulk:** 12 V max 2.5A (min 4.25 V) **Vin_mgmnt:** 3.3V max 110 mA)

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Xavier Rocher · just now

Author Developer

Memory	Signals
UDIMM	

Table 4 — Input/Output Functional Description (cont'd)

Symbol	Type	I/O Level	Function
HSCL	Input	1.0 V – 3.3 V	Host SidebandBus bus clock, supplied by the controller. Refer to JESD300-5 and JESD403-1
HSDA	Input / Output	1.0 V – 3.3 V	Host SidebandBus data, connected from the controller to Hub or Host bus Target devices. Refer to JESD300-5 and JESD403-1
HSA	Input	2.1 V max	Host SidebandBus bus device ID address pin; input to a Hub or other client device to distinguish between identical devices in the I3C-Basic address range.
RFU			Reserved for Future Use. No on DIMM electrical connection is present.
PWR_GOOD	Input / Output	Open Drain	Power good indicator. Open Drain output. The PMIC floats this pin high when VIN_Bulk input supply as well as all enabled output buck regulators and all LDO regulator tolerance threshold is maintained as configured in appropriate register. The PMIC drives this pin low when VIN_Bulk input goes below the threshold or when any of the enabled switch output regulators exceed the threshold configured in the appropriate register or any LDO output regulator exceeds the threshold tolerance. Input: The PMIC disables its output regulators when this pin is low. The LDO outputs shall remain on.
PWR_EN	Input	3.3 V	PMIC Enable. When this pin is high, the PMIC turns on the regulator. When this pin is low, the PMIC turns off the regulator. This signal is connected to PMIC's VR_EN pin.
VIN_BULK	Supply	5 V	5 V power input supply to the PMIC for analog circuits.
VSS	Supply		Ground

RDIMM

Table 6 — DDR5 RDIMM DC Operating Voltage

Symbol	Parameter	Voltage Rating (Volts)			Expected Current (Amps)[3]	Power State
		Minimum	Typical[4] (Nom)	Maximum		
VIN_BULK	Host Supply Voltage	4.25	12.0	15	2.5 (maximum)	Operational
VIN_MGMT	Host Supply Voltage	3.0	3.3	3.6	0.110 (maximum)	Operational
VDD[1]	PMIC Output Supply Voltage	1.067	1.1	1.166	Note 5	Operational
VDDQ[1]	PMIC Output Supply Voltage	1.067	1.1	1.166	Note 5	Operational
VPP[1]	PMIC Output Supply Voltage	1.746	1.8	1.098	Note 5	Operational
1.8V LDO[2]	PMIC Output Supply Voltage	Note 6	1.8	Note 6	0.025 (maximum)	Operational
1.0V LDO[2]	PMIC Output Supply Voltage	Note 6	1.0	Note 6	0.020 (maximum)	Operational

NOTE 1 The DDR5 SDRAM Standard must be met and takes precedence over this document.
NOTE 2 PMIC, Hub, TS, RCD Standards must be met and take precedence over this document.
NOTE 3 Maximum current establishes the platform maximum current regulation point. It provides a data point for DIMM developers to set power plane impedances.
NOTE 4 Typical voltage is platform dependent. This is a suggested value only.
NOTE 5 Maximum and Minimum Current ratings depend on PMIC 5000, 5010, and 5020 and number of SDRAM die placed.
NOTE 6 See PMIC supplier datasheets for Minimum and Maximum ratings.

SODIMM

Table 4 — Input/Output Functional Description (cont'd)

Symbol	Type	I/O Level	Function
HSCL	Input	1.0 V – 3.3 V	Host SidebandBus bus clock, supplied by the controller. Refer to JESD300-5 and JESD403-1
HSDA	Input / Output	1.0 V – 3.3 V	Host SidebandBus data, connected from the controller to Hub or Host bus Target devices. Refer to JESD300-5 and JESD403-1
HSA	Input	2.1 V max	Host SidebandBus bus device ID address pin; input to a Hub or other client device to distinguish between identical devices in the I3C-Basic/I2C address range.
RFU			Reserved for Future Use. No on DIMM electrical connection is present.
PWR_GOOD	Input / Output	Open Drain	Power good indicator. Open Drain output. The PMIC floats this pin high when VIN_Bulk input supply as well as all enabled output buck regulators and all LDO regulator tolerance threshold is maintained as configured in appropriate register. The PMIC drives this pin low when VIN_Bulk input goes below the threshold or when any of the enabled switch output regulators exceed the threshold configured in the appropriate register or any LDO output regulator exceeds the threshold tolerance. Input: The PMIC disables its output regulators when this pin is low. The LDO outputs shall remain on.
PWR_EN	Input	3.3 V	PMIC Enable. When this pin is high, the PMIC turns on the regulator. When this pin is low, the PMIC turns off the regulator. This signal is connected to PMIC's VR_EN pin.
VIN_BULK	Supply	5 V	5 V power input supply to the PMIC for analog circuits.
VSS	Supply		Ground

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Reply...

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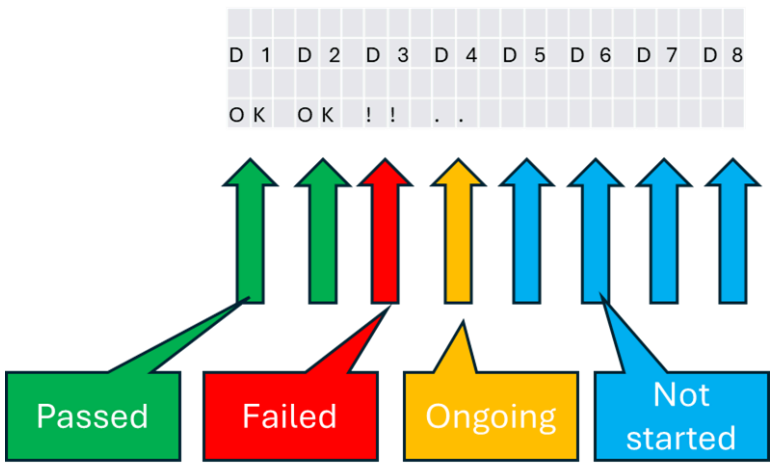
Leonardo Sala · just now Reporter

@Xavier.Rocher , as discussed during the meeting, I would prefer to simplify the HW in favor of a slightly more complex SW.

The status of the test can be monitored on the LCD itself, something like this:



Leonardo Sala · just now Reporter

@Xavier.Rocher , @Pedro.Silva ,

as we approach to final revision of the PV B, let's have a comprehensive design review to discuss all the aspecxts involved, with special focus on :

1. supply voltages (VDDIO, ...)
2. GPIO assignements
3. ...

Pedro Silva · 1 day ago Developer

Just as a reminder, pins PD6 (PWR_EN_RDIMM_6) and PD7 (PWR_EN_RDIMM_7) will be supplied by VDDIO2 and not VDD.

Check page 406 of this document: https://www.st.com/resource/en/reference_manual/rm0481-stm32h52333xx-stm32h56263xx-and-stm32h573xx-armbased-32bit-mcus-stmicroelectronics.pdf

Edited 1 day ago by Pedro Silva

Xavier Rocher · just now Author Developer

STM32CubeMx Configs

DDR5_TestPlatform.zip

NUCLEO_H563ZI_Final_Pinout.xlsx

Display I2C 50Khz

https://newhavendisplay.com/content/specs/NHD-0420D3Z-FL-GBW-V3.pdf

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